

Gender Differences in the levels of and Relationship between Competitive State Anxiety and Reaction Time for Hockey Players

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ABSTRACT

An athlete's mental state has long been thought to play a crucial role in his or her ability to perform sport-specific tasks. Players who are evenly matched in physical skills often rely upon their psychological skills to gain an advantage over their opponents. A complex and contrasting relationship has been highlighted between mood states, anxiety and motor performances. The present study aims to determine the level of somatic and cognitive anxiety and its effect on performances of Hockey players of different gender at junior level. A Total of 60 Hockey players, including 30 male and 30 female players, from the STC scheme of SAI, NS NIS, Patiala were selected for the study. The tools used were Reaction Time Apparatus (Chronoscope; RTM-608) and Competitive State Anxiety Inventory-2 (CSAI-2).

Gender-wise significant differences were found in the levels of subdomains of competitive state anxiety as well as reaction time and self-confidence. The relationship between competitive state anxiety and reaction time was found to be following the prominent theoretical conceptualizations for the combined sample. However, the relationship was found to be deviating significantly from the current viewpoints when gender was considered. Implications of the results and future directions are discussed in the main text.

Key words

Somatic Anxiety, Cognitive Anxiety, Reaction Time.

INTRODUCTION

In modern competitive sports, anxiety in sportsman has affected their performance as the physical load during the training of sportsman of international competition is also intensified. The Hockey players like other athletes are anxiety prone while participating in competitive sports

(Singh, 1982).

Spielberger (1966) published that anxiety can take two forms: state anxiety or trait anxiety. State anxiety refers to an emotional state consisting of fear or apprehension while trait anxiety refers to a predisposition to perceive situation as potentially threatening and respond with manifestations of state

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anxiety. State anxiety is "characterized by subjective, consciously perceived feelings of apprehension and tension, accompanied by or associated with activation or arousal of the autonomic nervous system" while trait anxiety is an "acquired behavioral disposition that predisposes an individual to perceive a wide range of objectively nondangerous circumstances as threatening and to respond to these with state anxiety reactions disproportionate in intensity to the magnitude of the objective danger". Athletes who are predisposed to higher levels of trait anxiety will perceive sport competition environments as being more threatening than they may actually be and respond with greater state anxiety responses.

An athlete's mental state has long been thought to play a crucial role in his or her ability to perform sport-specific tasks. Players who are evenly matched in physical skills often rely upon their psychological skills to gain an advantage over their opponents (Covassin & Pero, 2004). A complex and contrasting relationship has been highlighted between mood states, anxiety and motor performances (Vieira et al, 2006). In tasks with high information content, such as psychomotor ability or complex reaction time tasks, state anxiety can produce a slowing down of the information processing and motor strategies (Bolmont et al, 2000). However, in more simple tasks depending greatly on automatic abilities, state anxiety improves

attention (Pacheco-Unguetti et al, 2010) leading to better performances. Trait anxiety favors the processing of information of stimulus-response tasks, but not that of controlled tasks that require strategic processes (Bolmont et al, 2001). It has been shown that moderate state anxiety improves performance in visual and auditory response time both at the reaction time and movement time level (Hainaut & Bolmont, 2006). This agrees with previous data demonstrating that state anxiety increases arousal (Noteboom et al, 2001) and attention (Peretti, 1998), which can modulate sensory processing (Driver & Frackowiak, 2001).

State anxiety in sport can be manifested in many ways. Morris, Davis and Hutchings (1981) and Sarason (1984) indicated the existence of two distinct dimensions of state anxiety; cognitive anxiety; somatic anxiety. Cognitive anxiety is predominantly psychological in nature and is characterized by feelings of worry about outcomes and the use of negative mental imagery. Conversely, somatic anxiety is physiological in nature and includes increases in heart rate and increased perspiration. An excellent review of the relationship among forms of anxiety has recently been published (Smith, Smoll & Wiechman, 1998)

Multidimensional anxiety theory expanded on reversal theory's (Apter, 1982) inclusion of cognitive and physiological factors (Burton, 1988). In this model, cognitive anxiety (the central tenet of which is concerned with

the consequences of failure) has been found to have a negative linear relationship with performance (Burton, 1988). Self-confidence (a separate cognitive component) has been found to have a positive linear relationship with performance (Burton, 1988). Finally, somatic anxiety (physiological symptoms) has been found to have an inverted-U shaped relationship with performance (Burton, 1988). Although, this model incorporates many elements of anxiety, it still treats them as separate entities. The next model that arose looked at the interaction between two of these three factors.

The catastrophe model of anxiety and performance looks at the interactive effects of physiological arousal and cognitive anxiety upon performance (Fazey & Hardy, 1988; Hardy, 1990). Physiological arousal can influence performance as a result of the individual's interpretation of their physiological symptoms. According to the model as cognitive anxiety increases it will be beneficial to performance at low levels of physiological arousal but a detrimental effect at high levels of physiological arousal (Hardy et al, 1996). Furthermore, when cognitive anxiety is at a low level, changes in physiological arousal have little effect upon performance. However, as cognitive anxiety increases physiological arousal can have either a positive or negative effect on performance depending on how much arousal there is (Hardy et al, 1996). Once physiological arousal levels are

too high there is a steep drop in performance which can only be reversed by a reduction in physiological arousal (Hardy et al, 1996). Although the model fails to include a self-confidence variable, its interactive approach seems to be the best explanation for observed behavior.

However, because of lack of research on somatic and cognitive anxiety and its effect on performances, sports psychologists are still unable to determine the relationship among these variables. Moreover, most of the previous research has focused on elite athletes, ignoring less successful athletes. This was confirmed by Krane (1995) that research on competitive anxiety mainly focused on elite athletes. The extant literature also shows that there is a limited research comparing somatic and cognitive anxiety among Hockey athletes of different genders at junior level. The present study aims to determine the level of somatic and cognitive anxiety and its effect on performances between Hockey players of different gender at junior level. In other words, this research sought to find out the relationship between somatic & cognitive anxiety and performance.

METHODOLOGY

Design

Present study was a cross-sectional between groups study carried out using correlation design.

Sample

A Total of 60 Hockey players including 30 male and 30 female players

from the STC schemes of SAI, NS NIS, Patiala were selected for the study using purposive random sampling method. The age range was between 13 - 18 years and a minimum of 8th standard level of education was ensured. The sample consisted of mostly Hindu and Sikh participants from semi-urban background of Punjab and Haryana.

Tools

Reaction Time apparatus (Chronoscope; RTM-608):

Reaction time was recorded with the Multiple Choice Reaction Time Apparatus (Chronoscope; RTM – 608), a Medicaid Company product having resolution of 0.001 seconds. The visual stimuli are red, green and yellow lights. The auditory stimuli are three sounds having different pitches. The subjects sits to one side and examiner sits to other side of instrument. When examiner presses the relevant switches, visual and auditory stimuli appear on screen which is in front of the subjects and the instrument automatically starts counting of time. When subject press the key as a response to visual and auditory stimuli instrument stops counting of time. This time is taken as reaction time. During the reaction time testing, visual and auditory stimuli were presented for ten times for each of the reaction time categories. The visual and auditory reaction times were measured under two categories: 1. Simple Reaction Time. 2. Choice Reaction Time.

In the simple reaction time task subjects were tested for their visual and auditory reaction time by just reacting to

one particular light and one particular sound. In the choice visual and auditory reaction time tasks, the player reacts to three different colors of light i.e. red, green and yellow and three different pitches of sound by pressing the respective key for the experimentally presented color and the sound.

Competitive State Anxiety Inventory – 2 (CSAI-2):

The CSAI-2 (Martens et al, 1990) was used to measure intensity and directionality of anxiety interpretation, which includes subcomponents of cognitive anxiety, somatic anxiety and self-confidence. The intensity response scale consists of 27 items (nine in each subscale), rating intensity on a scale of one (not at all) to four (very much so), where overall scores could range from 9 (very low) to 36 (very high). Past research of the inventory has found to have strong internal consistency, with reliability coefficients of .80 and .90 for each subscale (Jones and Swain 1992). The instruction on the CSAI – 2 was based on how the athletes felt prior to the competition.

Procedure

After collecting demographic details of the athletes, necessary information about the study was provided and all the subjects were requested to provide their consent for participation in the study. Afterwards, the above mentioned questionnaire and equipment were used for data collection.

Statistical Analyses

Data thus obtained was analyzed

using the SPSS software. Chi-square test was used to compare the socio-demographic characteristics of the sample except age. Mean differences between genders on age and subdomains of competitive anxiety and reaction time were calculated using the t-test for independent samples. Pearson's correlation was calculated to

find out the relationship between the subdomains of competitive anxiety and reaction time for the total sample as well as males and females.

RESULTS & DISCUSSION

Results of the study are depicted and discussed in the following Tables and text:

Table -1: Socio demographic characteristics of the sample

Variables		Gender		Chi-square value	Sig.
		Male No. (%)	Female No. (%)		
Education	Up to 10 th	24 (80%)	18 (60%)	3.857	0.145
	Intermediate	5 (16.7%)	7 (23.3%)		
	Graduation	1 (3.7%)	5 (16.7%)		
Residential Background	Rural	15 (50%)	11 (36.7%)	4.615	0.099
	Semiurban	0	4 (13.3%)		
	Urban	15 (50%)	15 (50%)		
Religion	Hindu	14 (46.7%)	18 (60%)	1.833	0.400
	Sikh	15 (50%)	12 (40%)		
	Muslim	1 (3.3%)	0		
	Others	0	0		
Domicile	Punjab	14 (46.7%)	16 (53.3%)	3.633	0.458
	Haryana	13 (43.3%)	11 (36.7%)		
	Delhi	2 (6.7%)	0		
	U.P.	1 (3.3%)	2 (6.7%)		
	H.P.	0	1 (3.3%)		

*significant at $p < 0.05$ **significant at $p < 0.01$

Socio-demographic characteristics of the sample with respect to education, residential background, religion and domicile are depicted in Table 1. It shows that most of

the subjects were educated upto 10th i.e. highschool level and were from rural and urban residential backgrounds. It further shows that most of the participants were from Hindu and Sikh

communities of Punjab and Haryana. Results of chi-square test reveal that there was no significant difference found between the two groups for any of the socio-demographic variables. This shows that both groups were matched with respect to socio-demographic characteristics. Further, Table 2 shows that the mean age for male Hockey players was 14.73 ± 1.64 years and for female Hockey players it was 15.67 ± 2.23 years. Results of t-test shows that there was no significant difference between the two groups with respect to age also. Hence, the two groups were matched with respect to all the socio-demographic variables including age.

This Table also shows the socio-

demographic distribution of Hockey players at junior level in one of the most popular schemes i.e. SAI Training Centre (STC) at one of the premier sports training institutes i.e. Netaji Subhas National Institute of Sports (NSNIS), Patiala, Punjab, of Sports Authority of India (SAI). This scheme (STC) is being run by Sports Authority of India in most of the sports disciplines at a number of centers across country. The athletes selected under this scheme are sponsored by Government of India for sports training as well as education and accommodation. Athletes performing well under this scheme move to the next level of scheme called Center of Excellence (CoE) and then to the national teams of respective games.

Table-2: Comparison between the Study groups on Anxiety and Reaction Time

Variable	Male (M $\pm$ S.D.)	Female (M $\pm$ S.D.)	t-value	Sig.
Age	14.73 $\pm$ 1.64	15.67 $\pm$ 2.23	1.845	0.070
Cognitive State Anxiety	18.63 $\pm$ 5.01	20.27 $\pm$ 4.25	1.352	0.181
Somatic State Anxiety	16.00 $\pm$ 3.53	20.90 $\pm$ 4.28	4.832**	0.000
Self Confidence	30.87 $\pm$ 2.98	23.67 $\pm$ 5.27	6.516**	0.000
Simple Visual Reaction Time	342.5 $\pm$ 113.72	340.23 $\pm$ 190.45	.082	0.935
Simple Auditory Reaction Time	409.97 $\pm$ 206.85	354.23 $\pm$ 266.92	.978	0.332
Complex Visual Reaction Time	352.33 $\pm$ 53.93	417.00 $\pm$ 45.13	5.823**	0.000
Complex Auditory Reaction Time	618.2 $\pm$ 321.05	761.41 $\pm$ 368.81	1.864	0.067

*significant at $p < 0.05$ **significant at $p < 0.01$

Table 2 shows the results of comparison between the two groups on

study variables i.e. state anxiety and reaction time. As is evident from the

results, there was no significant difference found between the two groups on cognitive state anxiety. However, a significant difference at .01 level of significance was observed between males and females in the domains of somatic state anxiety and self-confidence. Important gender differences have been found by other researchers also focusing on the relationship between cognitive anxiety, self-confidence, and somatic anxiety. Results of a study done by Thuot, Kavouras and Kenefick (1998) also reveal that females had lower self-confidence and higher somatic anxiety scores than males on the CSAI-2. Thus, findings of the present study are supported by the findings of the previous studies.

This is partially supported by research that has focused on the determinants of anxiety as well as gender. Among males, cognitive and somatic anxiety was more strongly affected by their perception of opponent's ability and probability of winning (Jones, Swain & Cale, 1991). Female's cognitive anxiety and self-confidence is determined by readiness to perform and the importance they personally placed on doing well (Jones, Swain & Cale, 1991). These gender differences are indicative of the need to develop interventions that are tailored to individual needs and the importance of considering all factors when developing an intervention.

Results shown in Table 2 also reveal that the reaction times obtained for the simple tasks regardless of the

sensory modality or gender was shorter than the reaction time for the complex tasks. Our findings for the reaction times are parallel to the literature. The pioneer reaction time study was that of Donders (1868). He showed that a simple reaction time is shorter than a recognition reaction time, and that the choice reaction time is longest of all. This is in line with many studies concluding that a complex stimulus (e.g., several letters in symbol recognition vs. one letter) elicits a slower reaction time (Brebner & Welford, 1980; Folstein, Van Petten, Rose, 2008; Luce, 1986).

These results also show that males had a significantly faster reaction time than females for the complex tasks in almost both the sensory modalities. And if the range of the standard deviations obtained for both the groups are considered, this holds true in case of simple reaction time tasks too. However, the differences obtained are not statistically significant in case of simple reaction time tasks. These results are supported by a plethora of past studies, revealing that in every age group, males have faster reaction times than females, and female disadvantage is not reduced by practice (Dane and Erzurumluoglu, 2003; Der and Deary, 2006). The last study is remarkable because it included over 7400 subjects.

No significant difference between male and female athletes were found for simple reaction times in both the sensory modalities. In the past, a large number of studies have been done in different sports branches on different reaction times of athletes and different

results have been reported by Kosinski (2010). Our results may be due to the

similarity of the nature of athletics or differences in methods used.

Table-3: Correlation among Anxiety and Reaction Time for the total sample

Anxiety/Reaction Time	Simple Visual Reaction Time (r / sig.)	Simple Auditory Reaction Time (r / sig.)	Complex Visual Reaction Time (r / sig.)	Complex Auditory Reaction Time (r / sig.)
Cognitive State Anxiety	-.239 / .066	-.185 / .158	.073 / .577	.007 / .959
Somatic State Anxiety	-.038 / .774	-.144 / .271	.268**/ .038	.251 / .053
Self Confidence	.047 / .723	.112 / .396	-.317**/ .014	-.170 / .195

*significant at $p < 0.05$ **significant at $p < 0.01$

Table 3 shows the values and significance of correlation among the subdomains of anxiety and reaction time;. The Table shows that both cognitive and somatic state anxieties are negatively correlated with simple audio and visual reaction time, whereas, these are positively correlated with complex audio and visual reaction times. Moreover, the value of correlation among somatic state anxiety and complex audio and visual reaction time was also found to be statistically significant. Findings of the present study are supported by various studies done in past. A complex and contrasting relationship has been highlighted between anxiety and motor performances (Vieira et al, 2006). In tasks with high information content, such as psychomotor ability or complex reaction time tasks, state anxiety can produce a slowing down of the information processing and motor strategies (Bolmont et al, 2000). However, in more simple tasks depending greatly on automatic abilities, state anxiety improves

attention (Pacheco-Unguetti et al, 2010), leading to better performances. There are data suggesting that anxiety could favor, or at least not alter, the processing of information of relatively simple tasks, such as reaction time (Bolmont et al, 2000). Anxiety could favor the processing of information of stimulus-response tasks in antigenic conditions, but not that of complex motor skills requiring strategic processes (Bolmont et al, 2001). Other studies have shown that high anxiety facilitates performance in easy tasks and blocks performance in complex, difficult tasks (Etnyre & Kinugasa, 2002).

Cognitive anxiety has been found to have a negative linear relationship with performance (Burton, 1988). This is also evident from the results of the present study. For simple tasks there is a negative correlation among cognitive anxiety and reaction time indicating that increase in the level of cognitive anxiety is facilitative for performance of simple tasks in any of the two sensory modalities, whereas, the same is debilitating for the performance of complex tasks in both sensory

modalities. Self-confidence has been found to have a positive linear relationship with performance (Burton, 1988). In our study too, increase in the level of self-confidence is found to hinder the performance of simple tasks; whereas, the same is found to facilitate the performance of complex tasks in both sensory modalities. Finally, somatic anxiety has been found to have an inverted-U shaped relationship with performance (Burton, 1988). This is also evident from the results of the present study. For simple tasks; there is a negative correlation among somatic anxiety and reaction time indicating that increase in the level of somatic anxiety

is facilitative for performance of simple tasks in any of the two sensory modalities; whereas, the same is debilitating for the performance of complex tasks in both sensory modalities.

Another important finding to be noticed is the statistical significance of the relationship between somatic anxiety and complex reaction time in both sensory modalities. It shows that a significant increase in the level of somatic anxiety will sure cause a significant deterioration in the performance of complex task; whereas, the same cannot be said with confidence in case of cognitive anxiety.

Table-4: Correlation among Anxiety and Reaction Time for Male Players

Anxiety/Reaction Time	Simple Visual Reaction Time (r / sig.)	Simple Auditory Reaction Time (r / sig.)	Complex Visual Reaction Time (r / sig.)	Complex Auditory Reaction Time (r / sig.)
Cognitive State Anxiety	-.315 / .090	-.158 / .404	-.220 / .242	-.131 / .491
Somatic State Anxiety	-.403* / .027	-.472** / .008	-.095 / .618	.109 / .567
Self Confidence	.581** / .001	.406* / .026	.042 / .825	-.109 / .565

*significant at $p < 0.05$ **significant at $p < 0.01$

Table 4 shows the values and significance of correlation among subdomains of anxiety and reaction time for the male sample. It shows that the cognitive state anxiety was found to have a negative correlation with all the subdomains of reaction time but the values of the correlations were not statistically significant. This means that an increase in cognitive anxiety was found to have a facilitative effect on

performance of simple as well as complex tasks in both the sensory modalities for male sportsmen. In the case of somatic anxiety a statistically significant negative correlation was found for the simple task which shows that an increase in somatic anxiety is facilitative for the performance of simple tasks in both the sensory modalities. The correlation between somatic anxiety and complex reaction

time was found to be a very weak one and seems to follow the inverted U hypothesis (Burton, 1988). Self-confidence was found to a positive relationship with reaction time for most of the subdomains except complex

auditory reaction time. These findings are supported by the findings of the study done by Burton wherein it was found that Self-confidence has a positive linear relationship with performance (Burton, 1988).

Table-5: Correlation between Anxiety and Reaction Time for Female Players

Anxiety/Reaction Time	Simple Visual Reaction Time (r / sig.)	Simple Auditory Reaction Time (r / sig.)	Complex Visual Reaction Time (r / sig.)	Complex Auditory Reaction Time (r / sig.)
Cognitive State Anxiety	-.220 / .243	-.181 / .338	.212 / .260	.061 / .751
Somatic State Anxiety	.120 / .527	.153 / .421	-.079 / .678	.181 / .338
Self Confidence	-.102 / .592	-.123 / .519	.203 / .281	.022 / .907

*significant at $p < 0.05$ **significant at $p < 0.01$

Table 5 shows the values and significance of correlation among subdomains of anxiety and reaction time for the female sample. The Table shows that cognitive anxieties is negatively correlated with simple audio and visual reaction time; whereas, it is positively correlated with complex audio and visual reaction times. These findings prove the hypothesis that high anxiety facilitates performance in easy tasks and blocks performance in complex, difficult tasks (Etnyre & Kinugasa, 2002) for female athletes. The table also shows that somatic anxiety was found to be positively correlated with performance of all the subdomains of reaction time.

These findings are in complete contrast with the findings for the male sample where a negative correlation was found between somatic anxiety and most of the subdomains of reaction time. These findings indicate to the gender differences in the effect of anxiety on performance and should be taken care of while planning for intervention programmes meant for athletes of both genders. Further, an increase in the level of self-confidence is found to facilitate the performance of simple tasks; whereas, the same is found to hinder the performance of complex tasks in both sensory modalities for the female sample. This is again in complete contrast with the findings for the total sample where an inverse relationship

was found between the variables in consideration.

This again speaks very clearly of gender differences in anxiety performance relationship. Hence, the relationship between anxiety and sports performance is not as direct as has been thought of in past conceptualizations and consideration of gender of the athlete complicates the matter further. This calls for gender specific research in the field of sports in general and different sports in particular. The

consideration of type of sport poses another intriguing question. The mechanism through which the anxiety performance relationship works in case of individual versus team sports and combat versus non-combat sports with a special reference to gender of the athlete, needs to be explored diligently in future studies so that individualized intervention programmes could be developed as per the needs of the athletes, to help them with their performance.

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